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Fusion of short range radars and tailgate cameras for trailer angle estimation

Abstract

A method and system are disclosed for determining a trailer angle between a vehicle and a trailer attached to a the vehicle. The method includes receiving camera data from a camera supported by a rear portion of the vehicle, left radar data from a left short-range radar supported by a left rear portion of the vehicle, and right radar data from a right short-range radar supported by a right rear portion of the vehicle. A trailer angle is estimated based on the camera data, the left radar data, and the right radar data. The estimated trailer angle is sent to one or more vehicle systems associated with the vehicle.

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Background/Summary

(1) The present application is a National Stage Application under 35 U.S.C. § 371 of International Patent Application PCT/US2021/071055 filed on Jul. 29, 2021, and claims the benefit of U.S. provisional patent application No. 63/058,199, filed Jul. 29, 2020, in the U.S. Patent and Trademark Office which is incorporated by reference herein.

TECHNICAL FIELD

(1) This disclosure relates to a method and system for fusion short-range radar data and tailgate camera data for trailer angle estimation.

BACKGROUND

(2) Trailers are usually unpowered vehicles that are pulled by a powered tow vehicle. A trailer may be a utility trailer, a popup camper, a travel trailer, livestock trailer, flatbed trailer, enclosed car hauler, and boat trailer, among others. The tow vehicle may be a car, a crossover, a truck, a van, a sports-utility-vehicle (SUV), a recreational vehicle (RV), or any other vehicle configured to attach to the trailer and pull the trailer. The trailer may be attached to a powered vehicle using a trailer hitch. A receiver hitch mounts on the tow vehicle and connects to the trailer hitch to form a connection. The trailer hitch may be a ball and socket, a fifth wheel and gooseneck, or a trailer jack. Other attachment mechanisms may also be used.

(3) Some of the challenges that face tow vehicle drivers is driving in the rearward direction while being attached to the trailer because more than one person may be needed to maneuver the vehicle-trailer unit towards the specific location. Since the vehicle-trailer unit swivels around the hitch horizontally allowing the vehicle-trailer unit to move around corners, when the vehicle backs up, it pushes the trailer. Drivers are often confused as to which way to turn the vehicle steering wheel to get the desired change of direction of the trailer especially when backing into a parking spot. Applying an incorrect steering angle in the vehicle may also cause the trailer to jack-knife and lose its course. Therefore, oftentimes, one person drives the vehicle, e.g., the driver, and another one or more people are needed to view the tow vehicle and the trailer and provide the driver with directions regarding the path that the tow vehicle has to take. In some instances, the people providing directions to the driver may have difficulty providing efficient instructions for directing the path of the tow vehicle. Moreover, the driver may not be properly trained and/or may not have enough experience to back-up the tow vehicle and the trailer. In some examples, even trained drivers may struggle with backing up the tow vehicle while attached to the trailer because the trailer may have unfamiliar dimensions or is not easily maneuvered. Other difficulties may arise while parking the tow vehicle and the trailer, because the driver may have blind zones that do not allow adequate sight angles to properly park the tow vehicle and the trailer or the parking lot may be too tight to be able to maneuver the vehicle and the trailer.

(4) Some vehicles are equipped with a trailer reverse assist system that guides a trailer while backing up using a knob located inside of the vehicle. This is in lieu of the traditional way to back a trailer up, which is based on steering wheel input only. The trailer reverse assist function is meant

as a comfort feature to those inexperienced at backing a trailer. The trailer reverse assist system functions using a camera located in the tailgate of the vehicle which senses the trailer angle and wheelbase. In another implementation of the trailer reverse assist system, the estimated angle and wheelbase are provided to a driving function that controls the steering wheel angle and acceleration and braking are controlled by the tow vehicle driver.

(5) Some current systems only support traditional bumper mounted trailers and lack the ability to support fifth wheel and gooseneck trailers. In addition, there are limitations with current systems. Namely, the higher the angle made between the trailer and the vehicle, the more inaccurate the angle becomes through the tailgate camera. Some current implementations of trailer reverse assist systems may have trailer angle limits of 45 degrees. Additionally, the tailgate camera may not be able to support several classes of trailers, including fifth wheel and gooseneck trailers. Therefore, it is desirable to provide a system that overcomes the challenges faced by drivers of tow vehicles attached to a trailer.

SUMMARY

(6) Example embodiments of the present disclosure overcome shortcomings of current trailer related assist systems and thereby satisfy a need for a more robust estimation of the trailer angle formed between the tow vehicle and the trailer, for use by a trailer reverse assist system, for example. One example embodiment is directed to a method for determining at least one parameter associated with a trailer, such as a trailer angle between a trailer and an attached vehicle. The method includes receiving camera data from a camera supported by a rear portion of the vehicle, and receiving radar data from at least one short range radar supported by the vehicle. A trailer angle is estimated based on the camera data and the radar data. The estimated trailer angle is sent to one or more vehicle systems associated with the vehicle.

(7) Receiving the radar data may include receiving right radar data from a right short-range radar supported by a right rear portion of the vehicle, and receiving left radar data from a left short range radar supported by a left rear portion of the vehicle. In this instance, estimating the trailer angle is based upon the right radar data and the left radar data.

(8) The radar data may include an estimate of the trailer angle generated by the at least one short range radar, and the camera data may include an estimate of the trailer angle generated by the camera.

(9) Estimating the trailer angle may be performed by data processing hardware in the camera, in the at least one short range radar, or in the vehicle that is separate from the camera and the at least one short range radar.

(10) Estimating the trailer angle may use statistically weighted averaging of the radar data and the camera data. In this case, weights for the statistically weighted averaging are based upon one or more of a light condition of the environment of the vehicle and trailer, a most recently estimated trailer angle, or a number of reflections associated with the radar data. Estimating the trailer angle may use a Kalman filter. Estimating the trailer angle may include fusing the camera data and the radar data. In addition, the method may include estimating a wheelbase of the trailer based upon the camera data and the radar data, wherein the radar data includes an estimate of the wheelbase generated by the at least one short range radar, and the camera data includes an estimate of the wheelbase generated by the camera.

(11) An example embodiment is directed to a system for determining a trailer angle between a vehicle and a trailer attached thereto. The system includes data processing hardware and memory hardware in communication with the data processing hardware. The memory hardware stores instructions that when executed on the data processing hardware cause the data processing hardware to perform operations forming a method as described herein. The data processing hardware may be in the camera, in the at least one short range radar or in the vehicle that is separate from the camera and the at least one short range radar.

Description

DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a schematic view of an example vehicle-trailer system according to an example embodiment.
- (2) FIG. 2 is a schematic view of the exemplary vehicle-trailer system of FIG. 1 in which the trailer is at an oblique angle relative to the tow vehicle.
- (3) FIG. 3 is a schematic view of the tow vehicle of FIG. 1.
- (4) FIG. 4 is a schematic view of example vehicle-trailer system of FIG. 1 showing trailer angle estimations from different sensors.
- (5) FIG. 5 is a schematic view of an aspect of the vehicle-trailer system of FIG. 1 according to an example embodiment.
- (6) FIG. 6 is a schematic view of an aspect of the vehicle-trailer system of FIG. 1 according to another example embodiment.
- (7) Like reference symbols in the various drawings indicate like elements.

DETAILED DESCRIPTION

- (8) A tow vehicle, such as, but not limited to a car, a crossover, a truck, a semi-truck, a van, a sports-utility-vehicle (SUV), and a recreational vehicle (RV) may be configured to tow a trailer. The tow vehicle connects to the trailer by way of a trailer hitch. In general terms, example embodiments of the present disclosure are directed to a tow vehicle-trailer system which estimates the trailer angle between the tow vehicle and the connected trailer in part by fusing data from a rear-mounted camera and rear-mounted short range radars. The resulting trailer angle estimate, according to the example embodiments, allows for use with trailers including fifth wheel and gooseneck trailers. The trailer angle estimate may be used, for example, by a trailer reverse assist system for preventing a jack-knife event from occurring while the tow vehicle is operated in reverse.
- (9) Referring to FIGS. 1-3, in some implementations a vehicle-trailer system **100** includes a tow vehicle **102** attached to a trailer **104** by way of a hitch **106**. The tow vehicle **102** includes a drive system **110** associated with the tow vehicle **102** that maneuvers the tow vehicle **102** and thus the vehicle-trailer system **100** across a road surface based on drive commands having x, y, and z components, for example. As shown, the drive system **110** includes a front right wheel **112**, **112a**, a front left wheel **112**, **112b**, a rear right wheel **112**, **112c**, and a rear left wheel **112**, **112d**. In addition, the drive system **110** may include wheels (not shown) associated with the trailer **104**. The drive system **110** may include other wheel configurations as well. The drive system **110** includes other components (not shown) that are in communication with and connected to the wheels **112** and that allow the tow vehicle **102** to move, thus moving the trailer **104** as well. The drive system **110** may also include a brake system **114** that includes brakes (not shown) associated with each wheel **112**, **112a-d**, where each brake is associated with a wheel **112a-n** and is configured to slow down or stop the wheels **112a-n** from rotating. The tow vehicle **102** may also include an acceleration system **116** that is configured to adjust a speed and direction of the vehicle-trailer system **100**. The vehicle-trailer system **100** may include other systems as well.
- (10) The tow vehicle **102** may move across the road surface by various combinations of movements relative to three mutually perpendicular axes defined by the tow vehicle **102**: a transverse axis X.sub.V, a fore-aft axis Y.sub.V, and a central vertical axis Z.sub.V. The transverse axis X.sub.V extends between a right side R and a left side of the tow vehicle **102**. A forward drive direction along the fore-aft axis Y.sub.V is designated as F.sub.V, also referred to as a forward motion. In addition, an aft or rearward drive direction along the fore-aft direction Y.sub.V is designated as R.sub.V, also referred to as rearward motion. In some examples, the tow vehicle **102** includes a suspension system (not shown), which when adjusted causes the tow vehicle **102** to tilt

about the X.sub.V axis and/or the Y.sub.V axis, or move along the central vertical axis Z.sub.V. As the tow vehicle **102** moves, the trailer **104** follows along a path of the tow vehicle **102**. Therefore, when the tow vehicle **102** makes a turn as it moves in the forward direction F.sub.V, then the trailer **104** follows along.

(11) Moreover, the trailer **104** follows the tow vehicle **102** across the road surface by various combinations of movements relative to three mutually perpendicular axes defined by the trailer **104**: a trailer transverse axis X.sub.T, a trailer fore-aft axis Y.sub.T, and a trailer central vertical axis Z.sub.T. The trailer transverse axis X.sub.T extends between a right side R and a left side of the trailer **104**. A forward drive direction along the trailer fore-aft axis Y.sub.T is designated as F.sub.T, also referred to as a forward motion. In addition, a trailer aft or rearward drive direction along the fore-aft direction Y.sub.T is designated as R.sub.T, also referred to as rearward motion. Therefore, movement of the vehicle-trailer system **100** includes movement of the tow vehicle **102** along its transverse axis X.sub.V, fore-aft axis Y.sub.V, and central vertical axis Z.sub.V, and movement of the trailer **104** along its trailer transverse axis X.sub.T, trailer fore-aft axis Y.sub.T, and trailer central vertical axis Z.sub.T.

(12) The tow vehicle **102** may include a sensor system **120** to provide reliable and robust driving in a fully or partially autonomous manner. The sensor system **120** may include different types of sensors **122**, **124** that may be used separately or with one another to create a perception of the environment of the tow vehicle **102** that is used for the vehicle **102** to drive and aid the driver in make intelligent decisions based on objects and obstacles detected by the sensor system **120**. In some examples, the sensors **122**, **124** are supported by the tow vehicle **102** only, while in other examples, the trailer **104** may support one or more sensors **122**, **124**. The sensors **122**, **124** may be positioned on a rear section of the tow vehicle **102** for capturing a rearward environment of the tow vehicle **102**, a side section of the tow vehicle **102** for capturing a left/right environment of the tow vehicle **102**, and/or a front section of the tow vehicle **102** for capturing a front environment of the tow vehicle **102**. The combination of sensors **122**, **124** positioned around the tow vehicle **102** or the vehicle-trailer system **100** provides a surround view of the environment of the tow vehicle **102** and/or the trailer **104**.

(13) The sensor system **120** includes one or more cameras **122** that provide camera data **123**, **123a**. The one or more cameras **122** may include one or more monochrome and/or monocular cameras. As shown in FIGS. 1 and 2, the camera **122** is disposed along the rear of the tow vehicle **102**, such as along the rear bumper or the tailgate. It is understood that more than a single camera **122** may be used along the rear portion of the tow vehicle **102**. In some examples, the camera(s) **122** may include a fisheye lens that includes an ultra wide-angle lens that produces strong visual distortion intended to create a wide panoramic or hemispherical image **123a**. Fisheye cameras capture images **123a** having an extremely wide angle of view. Other types of cameras may also be used to capture images **123a** of the vehicle and trailer environment. The camera data **123** may include additional data **123b** such as intrinsic parameters (e.g., focal length, image sensor format, and principal point) and extrinsic parameters (e.g., the coordinate system transformations from 3D world coordinates to 3D camera coordinates, in other words, the extrinsic parameters define the position of the camera center and the heading of the camera in world coordinates). In addition, the camera data **123** may include minimum/maximum/average height of each camera **122** with respect to ground (e.g., when the vehicle is loaded and unloaded), and a longitudinal distance between the camera **122** and the tow vehicle hitch ball.

(14) The sensor system **120** includes short-range radar (SRR) **124** that provide radar data **125**. SRRs **124** operate around the vehicle **102** when in motion and detect potential impacts with objects and obstacles such as, but not limited to, cars, pedestrians, etc. In one embodiment, FIGS. 1 and 2 show a short range radar **124L** disposed along a left, rear portion of the tow vehicle **102** and a short range radar **124R** disposed along a right, rear portion of the tow vehicle **102**. It is understood that more or less than two short range radars may be disposed along the rear portion of the tow vehicle

102.

(15) In some examples, the sensor system **120** also includes one or more wheel encoders (not shown) associated with one or more wheels **112**, **112a-d** of the tow vehicle **102**. The sensor system **120** may also include one or more acceleration and wheel angle sensors (not shown) associated with the tow vehicle **102**. The acceleration and wheel angle sensors determine data that includes the acceleration of the tow vehicle **102** in the directions of the transverse axis X.sub.V and fore-aft axis Y.sub.V. The sensor system **120** may also include an IMU (inertial measurement unit) (not shown) configured to measure IMU data that includes the vehicle's linear acceleration (using one or more accelerometers) and rotational rate (using one or more gyroscopes). In some examples, the IMU data also includes a heading reference of the tow vehicle **102**. Therefore, the IMU data includes a pitch, roll, and yaw of the tow vehicle **102**.

(16) The sensor system **120** may include other sensors such as, but not limited to, radar, sonar, LIDAR (Light Detection and Ranging, which can entail optical remote sensing that measures properties of scattered light to find range and/or other information of a distant target), LADAR (Laser Detection and Ranging), ultrasonic sensors, stereo cameras, infrared sensor. etc. In some examples, the tow vehicle **102** includes a Global Positioning System (GPS) receiver (not shown) that determines a latitude and longitude position of the tow vehicle **102**. FIGS. **1** and **2** illustrate sensors **120** appearing along the front and side portions of the tow vehicle **102**, and such sensors may be cameras **122**, short range radars **124** and/or other sensors described above.

(17) The sensor data **121** may be used to alert the driver of possible obstacles when the vehicle-trailer system **100** is traveling in the forward direction F.sub.V or in the rearward direction R.sub.V, by way of audible alerts and/or visual alerts via the user interface **130**. Therefore, the sensor system **120** is especially useful for increasing safety in the vehicle-trailer system **100** which operates by the driver or under semi-autonomous or autonomous conditions.

(18) The tow vehicle **102** may include a user interface **130** that provides information to the driver and allows the driver to input commands. The user interface **130** receives one or more user commands from the driver via one or more input mechanisms, a touch screen display **132**, a microphone, and/or displays or outputs one or more notifications to the driver. In some examples, the user interface **130** includes a touch screen display **132**. In other examples, the user interface **130** is not a touchscreen and the driver may use an input device, such as, but not limited to, a rotary knob or a mouse for providing data from the driver or other user of the tow vehicle **102**.

(19) The drive system **110**, the sensor system **120**, and the user interface **130** are in communication with a vehicle controller **140**. The vehicle controller **140** includes a computing device (or data processing hardware) **142** (e.g., central processing unit having one or more computing processors) in communication with non-transitory memory or hardware memory **144** (e.g., a hard disk, flash memory, random-access memory) capable of storing instructions executable on the computing processor(s)). As shown, the vehicle controller **140** is supported by the tow vehicle **102**; however, the vehicle controller **140** may be separate from the tow vehicle **102** and in communication with the tow vehicle **102** via a network (not shown).

(20) In the example embodiments, the controller **140** includes a data fusion system or module **150** that analyses sensor system data **121** which includes camera data **123** and short-range radar data **125**. The data fusing system **150** fuses camera (image) data **123** from camera **122** with short range radar data **125** and estimates a trailer angle θ from the fused data, with the trailer angle being an angle formed between the fore-aft axis Y.sub.V of the tow vehicle **102** and the fore-aft axis Y.sub.T of the trailer **104**. The data fusion system **150** may form part of a trailer angle estimator system or module **152**. In one implementation, the trailer angle estimator system **152** forms part of a trailer reverse assist system or module **160** of the tow vehicle **102** for providing the estimated trailer angle θ ; and in another implementation illustrated in FIG. **3**, the trailer angle estimator system **152** is separate from and provides the estimated trailer angle to the trailer reverse assist system **160** and other vehicle systems which utilize trailer angle estimations. The inclusion of the short-range radar

data **125** in the data fusion system **150** enables use of additional classes of trailers **104** in the execution of a trailer reverse assist functions of the trailer reverse assist system **160** as well as other trailering functions performed by the tow vehicle **102**, for example. The trailer types that are enabled by the use of the data fusion system **150** include gooseneck trailers and fifth wheel trailers. A tow vehicle **102** having the data fusion system **150** adds robustness to the traditional sensor system **120** that is used by the trailer reverse assist system **160**, especially at higher trailer angles θ . (21) Because of the pivot point of fifth wheel trailers or gooseneck trailers, the towing vehicle **102** has the unique ability to have relatively higher trailer angles θ without causing a jack-knife event. Unfortunately, this means that the trailer **104** would be completely outside the field of view of the tailgate camera **122**. By adding short range radars **124** (**124L** and **124R**) to the rear corners of the vehicle **102** and using the short range radars to detect higher trailer angles θ between the trailer **104** and vehicle **102**, the data fusion system **150** enables gooseneck and fifth wheel trailers for use with the trailer reverse assist system **160**.

(22) In one example embodiment, each of the camera **122** and short range radars **124L**, **124R** independently estimates the trailer angle θ using the available raw data collected by the corresponding sensor. The short-range radars **124** estimate the trailer angle θ based upon the radar reflections from the trailer **104**. The calculated trailer angles θ from short range radars **124L** (i.e., the left short range radar) and **124R** (the right short range radar) are referred to as $\theta_{\text{sub.SRR_L}}$ and $\theta_{\text{sub.SRR_R}}$, respectively. The rear-mounted camera **122** estimates the trailer angle θ using the captured images **123** which is referred to as $\theta_{\text{sub.Camera}}$ as shown in FIGS. 5 and 6. Each individual estimate ($\theta_{\text{sub.SRR_L}}$, $\theta_{\text{sub.SRR_R}}$ and $\theta_{\text{sub.Camera}}$) is sent to a common processing block, in this case the data fusion system **150**, to perform data fusion and in particular to estimate the trailer angle $\theta_{\text{sub.Fused}}$ that is derived from the fused data. The estimated trailer angle $\theta_{\text{sub.Fused}}$ may be provided to and used by the trailer reverse assist system **160**, for example, for use in preventing a jack-knife event.

(23) The data fusion system **150** fuses the sensor data provided by camera **122** and short range radars **124L**, **124R** and generates a trailer angle estimate $\theta_{\text{sub.Fused}}$ based upon the sensor data received. In the illustrated example embodiment, the data fusing system **150** receives the trailer angle estimates from each of the camera **122** and the short range radars **124L** and **124R**, and performs statistically weighted averaging of the trailer angle estimates. The statistical weighed averaging assigns weights to each received trailer angle estimate θ based upon a level of trust in the received estimate. A number of factors are considered in setting the trust level and/or weight to each received sensor data (trailer angle estimate). Such factors may include the amount of light and/or the light conditions of the environment of the tow vehicle **102** and the trailer **104**, which affects the sensor data provided by the camera **122**; weather conditions in the environment of the tow vehicle **102** and the trailer **104**, including fog, which affects the sensor data provided by the camera **122**; the number of radar reflections which affects the sensor data provided by the short range radars **124L** and **124R**; and the last determined location of the trailer **104** relative to the tow vehicle **102**, which affects both the camera **122** and the short range radars **124L** and **124R**. The factors may be provided to the data fusion system **150** via sensors or other devices. It is understood that a number of other factors may be used in determining the weighting to apply to the sensor data provided by the camera **122** and the short range radars **124L** and **124R**. Based upon the provided factors, the data fusion system **150** determines weights to apply to the sensor data from each camera and short range radar, and estimates a trailer angle $\theta_{\text{sub.Fused}}$ that is based upon the received sensor data and the factors. The trailer angle estimate is then sent to, for example, the trailer reverse assist system **160** for use in preventing a jack-knife event from occurring.

(24) Information such as the wheelbase estimates and quality information may also be sent to improve the accuracy of the fused estimate of the trailer angle $\theta_{\text{sub.Fused}}$. The data fusion system includes a model which understands the benefits of using each sensor **122**, **124L** and **124R** at the various trailer angles θ in order to optimize the trailer angle estimate. For example, it is

advantageous to use the trailer angle estimates from the radars at higher angles as there is more trust in the short range radars **124** in high angle situations, so the data fusion system **150** gives greater weight to the trailer angle estimates $\theta_{\text{sub.SRR}}$ at higher trailer angles. Alternatively, the trailer angle estimates $\theta_{\text{sub.Camera}}$ from the camera **122** are more trustworthy at near zero angles and are given greater weights than trailer angle estimates from the short range radars **124L** and **124R**.

(25) In one implementation, the data fusing system **150** uses a Kalman filter to perform the statistical weighted averaging for estimating the trailer angle $\theta_{\text{sub.Fused}}$. It is understood that the statistical weighted averaging operation may have any of a number of different implementations.

(26) In the example embodiment(s) described above with respect to FIG. 5, each of the short range radar **124L** and **124R** and the camera **122** estimates the trailer angle θ based upon its own sensed data; and the data fusion system **150**, located in or executed by the controller **140** that is separate from the sensors **122**, **124**, receives the trailer angle estimates from each sensor **122**, **124** and fuses the trailer angle information for use in estimating the trailer angle $\theta_{\text{sub.Fused}}$. In another example embodiment, the camera **122** and the short range radars **124L** and **124R** do not estimate the trailer angle θ and instead provide raw sensor data to the data fusion system **150** of the controller **140**. In this example embodiment, the data fusion system **150** estimates trailer angles $\theta_{\text{sub.SRR_L}}$, $\theta_{\text{sub.SRR_R}}$, and $\theta_{\text{sub.camera}}$ in addition to estimating the trailer angle $\theta_{\text{sub.Fused}}$ therefrom.

(27) In yet another example embodiment, the data fusion system **150** forms part of the sensor system **120** and particularly one of the camera **122** or the short range radar **124L** and **124R** which has computational and/or processing capability for estimating trailer angle $\theta_{\text{sub.Fused}}$. In this example embodiment, the camera **122** or short range radar **124L**, **124R** having the data processing hardware which performs the trailer angle estimation receives sensor data from the camera **122** and short range radar **122**, **124** which does not estimate the trailer angle $\theta_{\text{sub.Fused}}$ as well as sensor data from the sensor which performs the trailer angle estimation.

(28) Fusing data between the short range radars **124L** and **124R** and the cameras **122** by the data fusion system **150** markedly improves the robustness of trailer assist systems of the two vehicle, especially in high angle situations and expands trailer assist functionality for use with gooseneck trailers and fifth wheels trailers. Further, the trailer angle estimator system **152**, which includes the data fusion system **150**, utilizes sensors commonly found on many vehicles for performing traditional radar functions such as blind spot detection, rear cross traffic alert, and trailer merge assist, so there is no additional hardware cost in including the trailer angle estimator system **152** in the tow vehicle **102**.

(29) In the example embodiment described above, the data fusion system **150** fuses trailer angle estimates from the rear facing camera **122** and the short range radars **124L** and **124R** to arrive at a fused trailer angle $\theta_{\text{sub.Fused}}$ therefrom. In another example embodiment, the data fusion system **150'** fuses the wheelbase WB of the trailer along with fusing the trailer angle θ . As shown in FIG. 6, each of the camera **122** and the short range radars **124L** and **124R** additionally estimates the wheelbase WB of the trailer **104**, i.e., the distance between the trailer's axles. The data fusion system **150'** receives the wheelbase estimate WB.sub.Camera from the camera **122**, the wheelbase estimate WB.sub.SRR_L from the short range radar **124L** and the wheelbase estimate WB.sub.SRR_R from the short range radar **124R**, and fuses the wheelbase estimates to determine the (fused) wheelbase WB.sub.Fused of the trailer **104**. According to this example embodiment, an implementation having a Kalman filter uses the same filtering algorithm for wheelbase and trailer angle determinations and outputs both the fused trailer angle $\theta_{\text{sub.Fused}}$ and the fused wheelbase WB.sub.Fused. The fused wheelbase WB.sub.Fused of the trailer **104** may be used in one or more trailer assist systems, such as the trailer reverse assist system **160**.

(30) Various implementations of the systems and techniques described here can be realized in digital electronic circuitry, integrated circuitry, specially designed ASICs (application specific integrated circuits), computer hardware, firmware, software, and/or combinations thereof. These

various implementations can include implementation in one or more computer programs that are executable and/or interpretable on a programmable system including at least one programmable processor, which may be special or general purpose, coupled to receive data and instructions from, and to transmit data and instructions to, a storage system, at least one input device, and at least one output device.

(31) These computer programs (also known as programs, software, software applications or code) include machine instructions for a programmable processor, and can be implemented in a high-level procedural and/or object-oriented programming language, and/or in assembly/machine language. As used herein, the terms “machine-readable medium” and “computer-readable medium” refer to any computer program product, apparatus and/or device (e.g., magnetic discs, optical disks, memory, Programmable Logic Devices (PLDs)) used to provide machine instructions and/or data to a programmable processor, including a machine-readable medium that receives machine instructions as a machine-readable signal. The term “machine-readable signal” refers to any signal used to provide machine instructions and/or data to a programmable processor.

(32) Implementations of the subject matter and the functional operations described in this specification can be implemented in digital electronic circuitry, or in computer software, firmware, or hardware, including the structures disclosed in this specification and their structural equivalents, or in combinations of one or more of them. Moreover, subject matter described in this specification can be implemented as one or more computer program products, i.e., one or more modules of computer program instructions encoded on a computer readable medium for execution by, or to control the operation of, data processing apparatus. The computer readable medium can be a machine-readable storage device, a machine-readable storage substrate, a memory device, a composition of matter effecting a machine-readable propagated signal, or a combination of one or more of them. The terms “data processing apparatus”, “computing device” and “computing processor” encompass all apparatus, devices, and machines for processing data, including by way of example a programmable processor, a computer, or multiple processors or computers. The apparatus can include, in addition to hardware, code that creates an execution environment for the computer program in question, e.g., code that constitutes processor firmware, a protocol stack, a database management system, an operating system, or a combination of one or more of them. A propagated signal is an artificially generated signal, e.g., a machine-generated electrical, optical, or electromagnetic signal that is generated to encode information for transmission to suitable receiver apparatus.

(33) Similarly, while operations are depicted in the drawings in a particular order, this should not be understood as requiring that such operations be performed in the particular order shown or in sequential order, or that all illustrated operations be performed, to achieve desirable results. In certain circumstances, multi-tasking and parallel processing may be advantageous. Moreover, the separation of various system components in the embodiments described above should not be understood as requiring such separation in all embodiments, and it should be understood that the described program components and systems can generally be integrated together in a single software product or packaged into multiple software products.

(34) A number of implementations have been described. Nevertheless, it will be understood that various modifications may be made without departing from the spirit and scope of the disclosure. Accordingly, other implementations are within the scope of the following claims.

Claims

1. A method for determining at least one parameter of a trailer coupled to a vehicle, the method comprising: receiving camera data from a camera supported by a rear portion of the vehicle; receiving radar data from at least one short range radar supported by the vehicle; estimating a trailer angle between a trailer and the vehicle coupled thereto based on the camera data and the

- radar data using statistically weighted averaging of the radar data and the camera data; and sending the estimated trailer angle to one or more vehicle systems associated with the vehicle.
2. The method of claim 1, wherein receiving radar data comprises receiving right radar data from a right short-range radar supported by a right rear portion of the vehicle, and receiving left radar data from a left short range radar supported by a left rear portion of the vehicle, and estimating the trailer angle is based upon the right radar data and the left radar data.
 3. The method of claim 1, wherein the radar data comprises an estimate of the trailer angle generated by the at least one short range radar, and the camera data comprises an estimate of the trailer angle generated by the camera.
 4. The method of claim 1, wherein estimating the trailer angle is performed by data processing hardware in the camera, in the at least one short range radar or in the vehicle that is separate from the camera and the at least one short range radar.
 5. The method of claim 1, wherein weights for the statistically weighted averaging are based upon one or more of a light condition of the environment of the vehicle and trailer, a most recently estimated trailer angle, or a number of reflections associated with the radar data.
 6. The method of claim 1, wherein estimating the trailer angle uses a Kalman filter.
 7. The method of claim 1, wherein estimating the trailer angle comprises fusing the camera data and the radar data.
 8. A method for determining at least one parameter of a trailer coupled to a vehicle, the method comprising: receiving camera data from a camera supported by a rear portion of the vehicle; receiving radar data from at least one short range radar supported by the vehicle; estimating a trailer angle between a trailer and the vehicle coupled thereto based on the camera data and the radar data; estimating a wheelbase of the trailer based upon the camera data and the radar data, wherein the radar data includes an estimate of the wheelbase generated by the at least one short range radar, and the camera data includes an estimate of the wheelbase generated by the camera; and sending at least one of the estimated trailer angle and estimated wheelbase to one or more vehicle systems associated with the vehicle.
 9. A system for determining at least one parameter of a trailer coupled to a vehicle, the system comprising: data processing hardware; and memory hardware in communication with the data processing hardware, the memory hardware storing instructions that when executed on the data processing hardware cause the data processing hardware to perform operations comprising: receiving camera data from a camera supported by a rear portion of the vehicle; receiving radar data from at least one short-range radar supported by a rear portion of the vehicle; estimating a trailer angle between a trailer and the vehicle coupled thereto based on the camera data and the right radar data using statistically weighted averaging of the left radar data the right radar data and the camera data; and sending the estimated trailer angle to one or more vehicle systems associated with the vehicle.
 10. The system of claim 9, wherein receiving radar data comprises receiving right radar data from a right short-range radar supported by a right rear portion of the vehicle, and receiving left radar data from a left short range radar supported by a left rear portion of the vehicle, and estimating the trailer angle is based upon the right radar data and the left radar data.
 11. The system of claim 9, wherein the data processing hardware is in the camera, in the at least one short range radar or in the vehicle that is separate from the camera and the at least one short range radar.
 12. The system of claim 9, wherein weights for the statistically weighted averaging are based upon one or more of a light condition of the environment of the vehicle and trailer, a most recently estimated trailer angle, or a number of reflections associated with the radar data.
 13. The system of claim 9, wherein estimating the trailer angle uses a Kalman filter.
 14. The system of claim 9, wherein estimating the trailer angle comprises fusing the camera data and the radar data.

15. The system of claim 9, wherein the radar data comprises an estimate of the trailer angle generated by the at least one short range radar and the camera data comprises an estimate of the trailer angle generated by the camera.

16. The system of claim 9, wherein the operations further comprises estimating a wheelbase of the trailer based upon the camera data and the radar data, wherein the radar data includes an estimate of the wheelbase generated by the at least one short range radar, and the camera data includes an estimate of the wheelbase generated by the camera.
